Sheharyar

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Education

University of Calgary, Schulich School of Engineering, Calgary, AB **May 2026**
B.Sc. Computer Engineering, With Distinction | GPA 3.86/4.0 (last two years)
Coursework: Machine Learning, Software Security, Software Performance Evaluation, Distributed Systems, Operating Systems

Experience

Riftyn | riftyn.com **Jan 2025 – Present**
Co-Founder and AI Engineer *Calgary, AB*
- Built an automated compliance auditor for construction permitting: tool-using LLM agents check submissions against building-code requirements and third-party records, then return schema-validated findings with confidence scores, source citations and an audit trail, while reviewers keep the final decision
- Built the evaluation harness that gates every release: labelled reviewer cases, versioned prompts and models, and regression tests that block a change when accuracy drops, with reviewer corrections folded into the next round
- Designed retrieval over PostgreSQL and pgvector and vision-language extraction from PDFs, and run the on-demand document pipeline on AWS at **99.5%** reliability; the system cut manual review effort **70%** in a customer pilot
- Work with coding agents daily (Claude Code with layered project context, memory files and custom commands), holding agent-written code to the same tests and review as my own

IP Factory, ChiselWare | with Rocksavage Technology Inc. and University of Calgary **Sep 2025 – Apr 2026**
Engineering Lead *Calgary, AB*
- Led a **6-person** team building a platform that lets defence and other mission-critical programs license verified hardware IP (written in Chisel, shipped as Verilog and SystemVerilog) instead of rebuilding it
- Built the verification pipeline that checks each generated IP configuration against its specification before release, with provenance tracking and **3**-tier access control, cutting validation cycles **50%**; presented results at **3** sponsor milestone reviews

Oil and Gas Development Company Limited (OGDCL) **Jul 2024 – Jan 2025**
Data and Machine Learning Engineering Intern *Remote*
- Built streaming anomaly detection on equipment telemetry that flags **15+** deviations per week for engineers, and Python and Spark ETL with validation and deduplication for an Oracle to SAP migration (**25%** faster)

Eurotecks | eurotecks.com **Jan 2024 – Present**
Software and Cloud Engineer *Remote*
- Traced a production slowdown to automated scraping, added rate limiting and TLS, and built monitoring across **3** services that cut outage detection time **80%**

Research and Technical Projects

Alberta Oil and Gas Value Map | Databricks hackathon, team project **Oct 2026**
Databricks, Spark SQL, Python, H3 geospatial indexing
- Turned **13M+** public production records into a municipal valuation of Alberta oil and gas after auditing the data: numeric fields stored as text, **25** activity types that double-counted output, and a mislabelled field we excluded from modelling
- Chose the prediction target with a leave-one-out test (Spearman $\rho = 0.78$) instead of assuming it, and validated a neighbour model with spatial block cross-validation, since random splits leak neighbouring cells (error **0.68** vs **0.90** for a regional baseline over **148** held-out blocks)
- Left cells with fewer than **3** observed neighbours unfilled rather than extrapolating, and mapped confidence next to every estimate

AWS Lambda Performance Study | Software Performance Evaluation, team project **Winter 2026**
Python, k6, AWS Lambda, CloudWatch
- Designed a reproducible study around **3** research questions, load-tested from **1** to **200** concurrent users, separated cold starts and throttling from true saturation, and reported P50/P95/P99 latency with 95% confidence intervals over repeated trials

Epidemic Forecasting with Calibrated Uncertainty
PyTorch
- Built a residual recurrent model with temporal attention and Monte Carlo dropout that produces 90-day forecasts with uncertainty bands across **51** countries

Competitions

- **Team Captain, International Young Physicists' Tournament** (Warsaw, 2019): led a national team that presented and defended original solutions to open-ended research problems against opposing teams.
- **CERN Beamline for Schools, team proposal** (2019, hosted at DESY): co-designed an experiment for the DESY II electron and positron test beam to test whether modulating beam angle and spin could carry information; one of **178** proposals from **49** countries.
- **National Champion, Pakistan Young Physicists' Tournament**: won the final with a mathematical model of fluid flow through small orifices.

Skills

AI and ML: LLM agents and tool use, evaluation harnesses, retrieval (RAG), structured outputs, Claude and OpenAI APIs, Claude Code, PyTorch, scikit-learn
Research Methods: experiment design, confidence intervals, cross-validation, rank correlation, load testing, failure analysis
Engineering: Python, TypeScript, SQL, C, C++; PostgreSQL, Databricks, Spark; Docker, Kubernetes, AWS, GitHub Actions; Chisel, SystemVerilog